

COLD CLIMATE HEAT PUMP OUTDOOR UNIT SCHEDULE - HPOU

UNIT ID	SERVICE	LOCATION	UNIT ELEC (VOLT/ PHASE)	UNIT ELEC (MCA)	MAX OVER CURRENT PROTECT ION (A)	NOMINAL COOLING TOTAL CAP. (KBH)	COOLING SEER2 (BTU/W/h)	COOLING SHR (SENS./ TOTAL)	HEATING COP @ 47° F OAT	HEATING COP @ 17° F OAT	HEATING COP @ 5° F OAT	ADJUSTED HEATING MAX. CAP. @ 3° F OAT (KBH)	HSPF2 (1V) (BTU/W h)	MANUF. OR EQUAL	MODEL OR EQUAL	REMARKS
HPOU-1	ATTIC	REAR, NE ELEV	240/1	29	48	36	15.2	0.76	3.2	1.7	1.8	34.4	9.1	MTSUBISHI	SUZ-AK36NLHZ	INSTALL SURGE PROTECTION DISCONNECT - SEE NARRATIVE
HPOU-2	ANNEX	REAR, NE ELEV	240/1	14	15	16	29.9	0.74	4.2	2.14	2.2	14.6	1.42	MTSUBISHI	MUZ-FX12NLHZ	INSTALL SURGE PROTECTION DISCONNECT - SEE NARRATIVE

COLD CLIMATE HEAT PUMP INDOOR UNIT SCHEDULE - HPIU

UNIT ID	LOCATION	SERVICE	AIR FLOW RATE (CFM)	EXT. STATIC SETTING (in WG)	FAN SPEED	UNIT VOLT/ PHASE	UNIT ELEC (MCA)	NOM. MAX. HEATING CAP @ 5°F OAT (KBH)	NOM. TOTAL CLG. CAP @ 95°F OAT (KBH)	*ADJ. HEATING CAP (KBH)	*ADJ. TOTAL COOLING CAP (KBH)	LIQ. LINE (IN)	GAS LINE (IN)	APPROX LINE LENGTH (FT)	MANUF. OR EQUAL	MODEL OR EQUAL	CONTROLS	REMARKS
		MAIN HOUSE FIRST FLOOR, ATTIC	870	0.5	MID	240/1		37	37	34.3	37.2	3/8	5/8	58	MTSUBISHI	SVZ-AP36NL	MHK2	R454B
HPIU-1	ATTIC				MID	240/1												
HPIU-2	ANNEX	ANNEX	270		MID	240/1	1	16.1	16.2	14.6	12.5	1/4	3/8	12	MTSUBISHI	MUZ-FX12NL	MHK2	R454B

*ADJUSTED VALUE FROM DIAMOND SYSTEM BUILDER SOFTWARE

ENERGY RECOVERY VENTILATOR SCHEDULE - ERV

UNIT ID	LOCATION	SERVICE	FLOW RATE (CFM)	EXTERNAL STATIC PRESSURE (IN. WG)	UNIT ELEC (VOLT/ PHASE)	UNIT ELEC (MCA)	MAX POWER (W)	SENS EFEC @ 32° F OAT (%)	MANUF. OR EQUAL	MODEL OR EQUAL	OPERATING WEIGHT (LBS)	ACCESSORIES	REMARKS
ERV-1	ATTIC	MAIN, ANNEX, BASEMENT, ATTIC	130	0.40	115/1	1.4	168	75	FANTECH	ATMO 200E	57	ECO-TOUCH AUTO IAQ, RTS-W BOOST TIMERS (2)	PROVIDE MERV-13 FILTER

DIFFUSER, REGISTER AND GRILLE SCHEDULE

ID	LISTED SIZE	AIR FLOW RATE RANGE (CFM)	THROW @ TV75 (FT)	FACE VELOCITY (FPM)	SERVICE	MOUNTING	ACCESSORI ES	MANUF / MODEL, OR EQUAL	REMARKS
SR-A	10" x 4"	100 - 140	9.5 - 13	500 - 700	SUPPLY AIR	SIDEWALL	VOLUME DAMPER	HART AND COOLEY / 618	STEEL, PAINTED
SR-B		141 - 220			SUPPLY AIR	SIDEWALL			
SR-C	10" x 8"	221 - 300	14.5 - 18	500 - 700	SUPPLY AIR	SIDEWALL	VOLUME DAMPER	HART AND COOLEY / 618	STEEL, PAINTED
SD-A	6" R	54 < x <= 95	3.5 - 4	< 700	SUPPLY AIR	CEILING	VOLUME DAMPER	HART AND COOLEY / 16	STEEL, PAINTED
SD-B	8" R	96 < x <= 160	5	<= 700	SUPPLY AIR	CEILING	VOLUME DAMPER	HART AND COOLEY / 16	STEEL, PAINTED
SD-C	10" R	161 < x <= 240	5	<= 700	SUPPLY AIR	CEILING	VOLUME DAMPER	HART AND COOLEY / 16	STEEL, PAINTED
SR-F1	4 x 10"	<= 60	10	300 - 400	SUPPLY AIR	FLOOR	VOLUME DAMPER	HART AND COOLEY / 210	STEEL, PAINTED
SR-F2	4" x 12"	60 < x <= 100	10	300 - 400	SUPPLY AIR	FLOOR	VOLUME DAMPER	HART AND COOLEY / 210	STEEL, PAINTED
V-A	4" R	< 50			VENTILATION SYSTEM SUPPLY AND RETURN VENTILATION	SIDEWALL OR CEILING		FANTECH / CG 4	
V-B	6" R	50 < x < 90			VENTILATION SYSTEM SUPPLY AND RETURN	SIDEWALL OR CEILING		FANTECH/ CG 6	
RG-1	12" x 12"	< 300		450	RETURN AIR	SIDEWALL OR CEILING		HART AND COOLEY / 650	STEEL, PAINTED
RG-2	12" x 16"	< 400		450	RETURN AIR	SIDEWALL OR CEILING		HART AND COOLEY / 650	STEEL, PAINTED
RG-3	20" x 20"	< 900		450	RETURN AIR	SIDEWALL OR CEILING		HART AND COOLEY / 650	STEEL, PAINTED

COORDINATE LOCATION AND SELECTION OF ALL FINISHES FOR DIFFUSER, REGISTERS AND GRILLES WITH THE OWNER

HEATING, VENTILATION AND AIR CONDITIONING SYSTEMS DESCRIPTION

SPACES			HEATING	COOLING	NOTES
BASEMENT			56 °F	65 °F	SPACE CONDITIONING INC FOR THIS PROJECT
FIRST FLOOR MAIN, ANNEX:			72 °F	74 °F	
ATTIC:			55 °F	65 °F	
OUTSIDE DESIGN CONDITIONS					
BASED ON FRAMINGHAM, MA (98.6% ASHRAE); OUTSIDE DESIGN CONDITIONS: HEATING DBT: 6 °F; COOLING DBT: 82 °F; WBTT: 71 °F (NOTE: COOLING DESIGN OAT: INCREASED 5 °F).					
CONDITIONED AREA (SF)					
THE CONDITIONED AREA IS THE FLOOR AREA WITHIN THE AIR BARRIER / THERMAL BOUNDARY, SUMMARIZED IN THE TABLE BELOW AND ATTACHMENT 1. EACH SPACE'S AREA IS A ROUNDED TO A MULTIPLE OF 5.					
VENTILATION RATE					
THE MINIMUM VENTILATION RATE, VIA STANDARD ASHRAE 62.2-2016, IS CALCULATED AS 0.03 CFM PER SQUARE FOOT TIMES THE CONDITIONED AREA PLUS THE NUMBER OF BEDROOMS PLUS 1 TIMES 7.5. OR, 0.03 x 3,310 + 7.5 x 4 => 130 CFM. ENERGY IS RECOVERED TO THE INCOMING AIR FROM THE EXHAUSTED AIR. THE STANDARDS' INFILTRATION CREDIT IS NOT ASSUMED.					
HEATING AND AIR CONDITIONING SYSTEMS - CENTRAL HEAT PUMPS AIR SYSTEMS					
THE MAIN HOUSE'S FIRST FLOOR'S PRIMARY HEATING AND COOLING SHALL BE A SINGLE ZONE COLDCIMATE RATED HEAT PUMP SYSTEM. HP-1, INCLUDING WITH INDOOR AIR HANDLER. THE AIR HANDLER SHALL BE LOCATED IN THE ATTIC. - SEE H-4. - THE ANNEX'S PRIMARY HEATING AND COOLING SYSTEM SHALL BE A COLDCIMATE, SINGLE ZONE, HIGH WALL, HEAT PUMP SYSTEM. HP-2. THE BUILDING'S OIL-FIRED FORCED HOT WATER SYSTEM SHALL BE RETAINED FOR EMERGENCY HEATING AND DOMESTIC WATER SERVICE.					
AIR FLOW CONCCEPT					
A VOLUME FLOW RATE (CFM) OF CONDITIONED AIR SHALL BE SUPPLIED TO EACH MAJOR SPACE PROPORTIONAL TO THE HEATING OR COOLING LOAD. - CONDITIONING AIR WILL RETURN TO ITS SOURCE THROUGH COMMON SPACES. - THE HP-1 SYSTEM'S MAIN AIR RETURN IS IN THE SIDEWALL OF THE KITCHEN SKYLIGHT. A SECONDARY RETURN IS LOCATED IN THE PRIMARY BEDROOM.					
VENTILATION SYSTEM - INDOOR AIR QUALITY					
VENTILATION (INCOMING OUTDOOR AIR) OR EXHAUST AIR MUST FLOW THROUGH ALL SPACES CONSTANTLY TO MAINTAIN GOOD INDOOR AIR QUALITY. THE RESIDENCES VENTILATION AND EXHAUST SHALL BE PROVIDED BY AN ENERGY RECOVERY VENTILATOR, ERV-1, AND ASSOCIATED DUCT SYSTEM. THE CONCEPT IS TO SUPPLY FRESH AIR AT THE PERIMETER OF OCCUPIED SPACES, SUCH AS LIVING AREA AND BEDROOMS, AND EXHAUST THROUGH BATHROOMS AND UTILITY AREAS. IT IS IMPORTANT FOR ALL SPACES TO BE EITHER SUPPLIED OR EXHAUSTED TO AVOID MUSTY AND DOOR ACCUMULATION.					
THE ERV SHOULD OPERATE AT ITS BACKGROUND FLOW RATE WHENEVER THE RESIDENCE IS OCCUPIED AND ITS WINDOWS ARE CLOSED.					
KITCHEN RANGE HOOD EXHAUST SYSTEM					
THE EXISTING KITCHEN RANGE HOOD EXHAUST SYSTEM SHALL BE RETROFITTED WITH AN EXTERIOR FAN, ATTIC DUCTWORK AND A MODULATING WALL CONTROL.					
HVAC SYSTEMS CONTROLS					
HP-1 AND HP-2 SHALL BE CONTROLLED BY WALL-MOUNTED THERMOSTATS, MITSUBISHI MHK-2, OR EQUAL. ALSO, HP-2 MAY BE CONTROLLED WITH ITS HAND-HELD REMOTE CONTROLLER. THE HEAT PUMP SYSTEMS SHALL BE OPERATED AS THE PRIMARY MEANS OF SPACE HEATING AND AIR CONDITIONING.					
ERV-1 SHALL OPERATE IN ONE OF TWO MODES, EITHER THE DESIGN FLOW RATE, CONTINUOUSLY, OR A HIGHER FLOW RATE, "BOOST" MODE, INTERMITTENTLY. PER OCCUPANT DEMAND. THE BOOST MODE IS INTENDED FOR BATHROOM EXHAUST AND CAN BE INITIATED BY A TIMER SWITCH. A CENTRAL CONTROLLER SHALL ALLOW AN OCCUPANT TO OPERATE AND MONITOR THE SYSTEM.					
GENERAL CONSIDERATIONS					
THE DRAWINGS ARE NECESSARILY DIAGRAMMATIC AND DO NOT SHOW ACTUAL LOCATIONS. THE HVAC SUBCONTRACTOR SHALL DETERMINE AND COORDINATE THE ACTUAL LOCATIONS OF EQUIPMENT, TERMINALS, AND ELEVATIONS OF ALL EQUIPMENT WITH THE OWNERS AND OTHER SUBCONTRACTORS. VENT TERMINAL LOCATIONS AND DEVICES (GRILLES, DIFFUSERS, ETC.) SHALL BE APPROVED BY THE OWNERS BEFORE INSTALLATION.					
REFRIGERANT TUBING, CONDENSATE PIPING, DUCTWORK, ELECTRIC CONDUIT, ETC. SHALL BE INSTALLED NEATLY AND GENERALLY PARALLEL TO THE BUILDING STRUCTURE.					
VIBRATION AND NOISE FROM OPERATING ATTIC HVAC EQUIPMENT SHALL BE PREVENTED FROM TRANSMISSION TO THE ATTIC FLOOR. IT MAY BE SUSPENDED FROM THE ROOF RAFTERS OR, OTHERWISE, ISOLATED FROM THE ATTIC FLOOR. OWNER SHALL APPROVE THE MEANS OF ISOLATION.					
SEE DRAWING H-1 FOR A SCHEDULE OF PROPOSED EQUIPMENT. EQUIPMENT MUST BE QUOTED AS LISTED. HOWEVER, ALTERNATE EQUIPMENT OR SYSTEMS MAY BE QUOTED SEPARATELY AND OFFERED TO THE OWNERS. APPLICATIONS FOR MASSAVE INCENTIVES SHALL BE COMPLETED ON BEHALF OF THE OWNER.					
THE SYSTEM INSTALLATION SHALL COMPLY WITH ALL MANUFACTURER REQUIREMENTS. A SURGE PROTECTION DISCONNECT SHALL BE INSTALLED FOR EACH OUTDOOR UNIT. MARS 8391 IS, OR EQUAL.					
THE HEAT PUMP AND ERV SYSTEMS SHALL BE COMMISSIONED. AIR FLOW RATE AND STATIC PRESSURE MEASUREMENTS SHALL BE RECORDED ON A SCHEMATIC OF THE SYSTEM. RESULTS SHALL BE REPORTED TO THE OWNER.					
THE FOLLOWING COMMISSIONING ITEMS SHALL BE PERFORMED:					
INDOOR UNITS					
INSTALLED LEVEL, SECURE, ISOLATED FROM VIBRATION					
DUCTWORK SEALED/INSULATED; CONDENSATE DRAIN TESTED					
ELECTRICAL CONNECTIONS TIGHT AND CORRECT					
FACTORY FILTERS CLEANED AND REINSTALLED					
EXTERNAL STATIC PRESSURE SETTING (0.3, 0.5, 0.8)					
ESP MEASURED AND NOTED					
AIR FLOWS BALANCED AND TOTAL CFM VERIFIED					
THERMOSTAT/CONTROL, WIRED AND CONFIGURED					
COMMUNICATION WITH OUTDOOR UNIT CONFIRMED					
SAFETY INTERLOCKS VERIFIED					
BLOWER OPERATION IN HEATING/COOLING CONFIRMED					
BLOWER SPEED/FAN PERFORMANCE PER DESIGN					
DEHUMIDIFICATION/SUPPLEMENTAL FUNCTIONS TESTED					
SUPPLY/RETURN TEMPERATURES RECORDED					
SYSTEM CYCLES/RESPONDS TO THERMOSTAT					
FINAL TEST DATA RECORDED AND O&M INSTRUCTIONS PROVIDED					
OUTDOOR UNITS					
INSTALLED LEVEL, ON PAD/BRACKETS WITH CLEARANCES					
REFRIGERANT LINES INSULATED, SUPPORTED, LEAK-CHECKED, EVACUATED					
SERVICE VALVES OPEN AND CONNECTIONS TIGHT					
ELECTRICAL DISCONNECT, GROUNDING, AND PROTECTION CORRECT					
AIRFLOW CLEARANCES FREE OF OBSTRUCTIONS					
FAN AND COIL SURFACES CLEAN					
CONTROL WIRING CORRECT AND IDENTIFIED					
SYSTEM RESPONDS TO THERMOSTAT/CONTROL					
COMPRESSOR AND FAN START SMOOTHLY					
REFRIGERANT CHARGE VERIFIED (PER MANUFACTURER METHOD)					
PRESSURES, TEMPS, VOLTAGE AND AMPERAGE WITHIN SPEC					
OUTDOOR AMBIENT TEMPERATURE RECORDED					
DISCONNECT LABELED AND MAINTENANCE REVIEWED WITH OWNER					
FINAL OPERATING DATA RECORDED					
OWNERS		Vander Straeten, Ghosh			
HVAC Design		Wayland, MA			
OCTOBER ENGINEERING LLC					
Sudbury, MA 01776					
508-561-7553					
ENERGY RETROFIT AND ELECTRIFICATION					
13 Rice Spring Lane					
Wayland, MA 01778					
SCHEDULES, NARRATIVE					
10-10-2025		H-2			
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